

Krishnkant (Krish) Swarnkar

Senior AI Research Engineer building production grade Agentic AI systems for financial intelligence; Experience in system design, deep learning (model training and deploying) and model evaluations (quality and reliability).

Education

University of Massachusetts, Amherst — MS, Computer Science (Research Track, AI) Sep 2019 - May 2021
GPA: 4.0/4.0 | Selected coursework: Reinforcement Learning, Probabilistic Graphical Models, Statistical Inference
IIT (Indian Institute of Technology) BHU — BTech, Computer Science (Honors) Jul 2015 - May 2019
CPI: 9.14/10

Core Skills

AI Agent Development, Harness, LLMs, RAG, Embeddings, Reranking, Summarization, Extraction, Classification, NLU
| PyTorch, Transformers, LangChain, Agent SDK | Python, FastAPI, Prometheus, Grafana, Docker, Kubernetes

Industry Experience

AlphaSense Inc. Jun 2021 - Present
Senior AI Engineer New York, NY

- Developed **Autonomous Agents operating in sandbox environments** to generate well-cited and grounded pptx slide decks that match institutional decks in layouts and structure.
- Developed key components of a **Market Research AI Agent, powered by RAG**, including planning/ orchestration, query understanding, tools (search and analytics), and citation grounding, serving over 20k queries daily. Optimized avg TTFT to 4.5s while maintaining quality through LLM based planning, query routing, and efficient context handling.
- Built automated summarization modules** for Expert Call summaries and Earnings Call highlights, lowlights, and trending topics summaries using RAG and prompt engineering. **Trained and deployed a sentiment analysis guardrail** for LLaMA2-70B summaries with 99%+ accuracy, and reduced hallucinations via context-aware chunking.
- Designed and evaluated a **knowledge extraction pipeline** to power an **M&A Deal Screener** (buyer, target, price, date, etc) from press and news, guiding data annotation, and conducting **NER experiments** with BERT and LLMs. Maintained legacy Boilerplate Detection and Themes Extraction systems alongside for downstream content discovery.
- Designed and deployed an Earnings Calls **KPI Extraction** system (quantized Electra model), improving Amount-KPI accuracy by 18.7% (F1 by 14.4%) over a rule-based baseline, enabling reliable KPI ingestion for downstream analytics.

Robert Bosch LLC May 2020 - Aug 2020
NLU Research Intern Sunnyvale, CA

- Implemented slot filling and intent detection models in PyTorch for NLU, and proposed a novel **BiLSTM Attention architecture** delivering +2% absolute F1 over the prior state of the art, and implemented its **extension to BERT**.

Publications

- Text Processing Like Humans Do: Visually Attacking and Shielding NLP Systems. **NAACL-HLT 2019**. [DOI/TODO]
- Improved Cross-Lingual Question Retrieval for Community Question Answering. **WWW 2019**. [DOI/TODO]
- Di-LSTM Contrast: A Deep Neural Network for Metaphor Detection. **FigLang (NAACL) 2018**. [DOI/TODO]

Academic Research Experience

Information Extraction and Synthesis Lab (IESL), UMass Amherst *Research Project* Jan 2020 - May 2021

- Reduced stereotypical bias (gender, racial, political) in contextualized embeddings from ELMo and BERT. *Advisor: Prof. Andrew McCallum.*

Ubiquitous Knowledge Processing Lab (UKP), TU Darmstadt *Research Intern* May 2018 - Nov 2018

- Demonstrated visually inspired adversarial attacks on NLP models and proposed ways to improve robustness; extended cross-lingual (German to English) question retrieval to improve translation robustness. *Advisor: Prof. Iryna Gurevych.*

Natural Language Processing Research Lab, IIT (BHU) *Research Intern* May 2017 - Jul 2017

- Built a pipeline-based QA system converting unstructured text into structured representations and answering queries using graph matching, dependency relations, and rules. *Advisor: Prof. Anil Kumar Singh, Prof. Rajeev Sangal*

Awards / Service

- Reviewer, ACL 2020 and NAACL 2018 FigLang Workshop
- Honourable Mention, ACM-ICPC Regionals